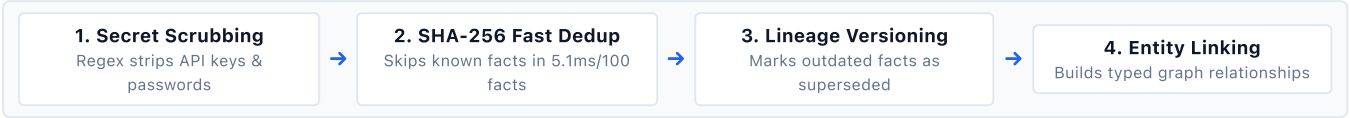


Sub-Millisecond Knowledge Ingestion

The 4-Stage Absorb Pipeline: Deduplication, Lineage Chains & Secret Redaction

The 4-Stage Memory Absorb Pipeline (94 μs / Fact)

Raw information from web searches and research sessions cannot be dumped blindly into memory. Fathom processes facts through a strict, multi-stage curation pipeline in under **94 microseconds per fact**



APPEND-ONLY TRUTH & LINEAGE CHAINS

Facts are never silently overwritten. If a company raises a Series B round after previously raising Series A, the old fact is marked with a **supersedes** edge, preserving the historical evolution of truth for auditing.

HYBRID RETRIEVAL: 0.70 VECTOR + 0.30 BM25

Queries combine the exact keyword precision of SQLite FTS5 (for names, emails, and codes) with semantic vector similarity, delivering **sub-2ms search latencies** across thousands of memories.

AUTOMATED GARBAGE COLLECTION (GC)

A background pruning routine cleans up ephemeral task artifacts (configurable 30-day TTL) while permanently retaining core entity relationships and verified client preferences.

Deterministic Fact Compaction: The knowledge base stays lean and focused on high-signal business facts without accumulating noisy search fragments.